Software Design Description 010

LoRa Calculations

Set Baud : 115200

NOTE: This does not work how one would think "normal UART" works.

ROI	Order of events	Final event
Transmitter	ESP32-S3 TX -> REYAX RX AT+SEND=...	Command to send
Transmitter	REYAX TX -> ESP32-S3 RX +OK	Acknowledgment
Receiver	REYAX TX -> DevKitV1 RX +RCV=...	Incoming data notification

This took me longer than it should have to figure out.

Spreading Factor Set for 2 Miles

Spreading factor	Receiver sensitivity for bandwidth fixed at 125 kHz
SF7	-123 dBm
SF8	-126 dBm
SF9	-129 dBm
SF10	-132 dBm
SF11	-134.5 dBm
SF12	-137 dBm

REYAX RYLR Specs:

$$f = 915MHz = 915 \times 10^6$$

$$T_x = +13dBm(power)$$

$$gain = \ +2dB_i$$

$$D : 2miles = \ 3.22Km$$

$$C : 299792458\frac{m}{s}$$

Formula: free space path loss

$$FSPL(dB) = 20(log_{10}(d) + log_{10}(f) + log_{10}(\frac{4\pi}{C})) - (G_{T_x} - G_{R_x})$$

$$= 20log_{10}(d \times f \times \frac{4\pi}{C}) = 20log_{10}(3200 \times 915 \times 10^6 \times \frac{4\pi}{C}) - (4)$$

$$= 97.7792dB$$

This assuming ~15 dB of loss

$$P_R = +13 - 97.7792dB - 15 = -99.792$$

SF7	-123 dBm	23.22
SF8	-126 dBm	26.22
SF9	-129 dBm	29.2203

Bit Rates:

$$\Delta f = 125kHz \rightarrow X = SF \times \frac{\Delta f}{2^{SF}} \times R_c$$

$$R_c = \frac{4}{5}$$

This is assuming that for every 4 bits theres odd 1 parity.

Spread factor	Bitrate (X)
SF7	5468.75 b/s
SF8	3125 b/s
SF9	1757.81 b/s

SF7 should provide adequate range. If this turns out to be functionally false a dynamic range will need to be implemented.